

Sonifying the Heart: Turning ECG Data into Music

Music Workshop Sprint 1: Initial Project Proposal

Chervith Reddy Manidhar Sukasi Sathwik Reddy

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1. The Idea

We want to build a system that converts real electrocardiogram (ECG) recordings into music, so that a healthy heart rhythm and a diseased one become *audibly* different.

The heartbeat already supplies a natural rhythm: the time between successive beats (the R–R interval) determines note onsets, beat-to-beat variability controls how consonant or dissonant the harmony becomes, and signal amplitude controls dynamics. A normal sinus rhythm should therefore produce a steady, consonant musical pattern, whereas atrial fibrillation should sound more irregular and unstable.

2. Why This Idea

- It sits naturally at the intersection of healthcare and music; the musical structure is not merely decorative, but carries the underlying physiological information.
- It has potential applications in teaching arrhythmia recognition, accessibility for visually impaired clinicians, and eyes-free physiological monitoring.

3. Rough Approach

- **Data.** We will use the MIT-BIH Arrhythmia Database (`mitdb`) and MIT-BIH Atrial Fibrillation Database (`afdb`) from PhysioNet. Both are openly available. Crucially, `afdb` provides pre-computed R-peak positions and cardiologist-verified rhythm labels, so beat detection does not need to be developed from scratch.
- **Extract.** We will read records using the `wfdb` Python package and compute R–R interval series and variability measures such as RMSSD. Segments will then be selected according to rhythm labels, primarily normal sinus rhythm versus atrial fibrillation.
- **Map.** We will design a sonification mapping in which R–R intervals determine note onsets and durations, local deviation from a running mean controls pitch selection, and ECG amplitude controls note velocity. Stable rhythms can remain within a pentatonic scale, while increasing irregularity can introduce chromatic pitch movement.
- **Render.** We will generate MIDI using `pretty_midi` or `mido` and render it using FluidSynth or a digital audio workstation. The goal is to produce short paired clips that make the contrast between rhythms immediately audible.
- **Present.** We will present a small collection of audio examples together with plots aligning the ECG waveform with the generated musical score, accompanied by a short explanation of the mapping rationale.

4. Tools

The implementation will primarily use Python, with `wfdb` for PhysioNet data access, `numpy` and `scipy` for signal processing, `pretty_midi` or `mido` for MIDI generation, FluidSynth or a DAW

for audio rendering, and `matplotlib` for visualisation.

An LLM-assisted layer may optionally be explored to suggest alternative musical mappings, although this will not be central to the project.

5. Scope and Related Work

ECG sonification is an established research area, and recent work in *npj Digital Medicine* has highlighted the need for reproducible reference implementations on shared benchmarks.